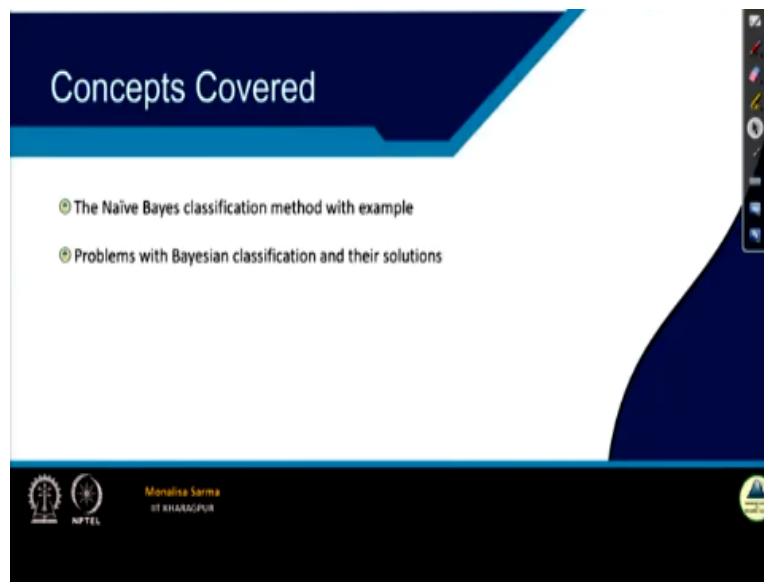


Statistical Learning for Reliability Analysis
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Lecture - 51
Bayesian Classification (Part-II)

Hi guys, so in continuation of our earlier lecture on Bayesian classification so we will be continuing from what way we have ended last lecture.

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So in today's lecture we will see the name based classification method we will see that within help of an example and base classification it has some problems as well as the solution also exists for that. So we will be discussing those problems as well as the solutions.

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Bayesian Classifier

- Suppose, Y is a class variable and $X = \{X_1, X_2, \dots, X_n\}$ is a set of attributes, with instance of Y .

INPUT (X)	CLASS(Y)
...	...
...	...
$X_1, X_2, \dots, X_n$	Y_i
...	...

- The classification problem, then can be expressed as the class-conditional probability

$$P(Y = y_i | (X_1 = x_1) \text{ AND } (X_2 = x_2) \text{ AND } \dots (X_n = x_n))$$

So now so first is base classification classifier what is it that it is the what to say based on the base theorem we have seen how is it right. So now just a formal if I can write suppose Y is a class variable and X is equals to this x is the set of attributes, so if Y is a class variable means y_1, y_2, y_3 whatever now if there are four classes so y_1, y_2, y_3, y_4 the classes. So I am using variable Y for the classes and variable X for the set of attributes where X consists of;